

Fundamentals of Operating System Design

Module 6: User Mode and Kernel Mode

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Learning Outcomes

- Explain user mode and kernel mode.
- Understand why CPUs provide two execution modes.
- Describe privileged instructions.
- Explain mode switching using system calls and interrupts.

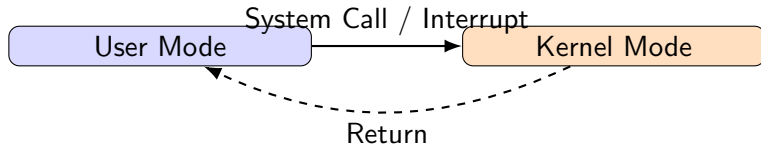
Why Two CPU Modes?

Need

Modern operating systems protect hardware and critical data by executing applications and the operating system in different CPU modes.

- Prevent accidental damage
- Prevent unauthorized hardware access
- Improve system stability
- Improve security

User Mode vs Kernel Mode



Characteristics of User Mode

- Applications execute here.
- Cannot execute privileged instructions.
- Cannot directly access hardware.
- Limited permissions.
- Safer execution environment.

Characteristics of Kernel Mode

- Operating system kernel executes here.
- Full access to hardware.
- Executes privileged instructions.
- Handles interrupts and exceptions.
- Manages CPU, memory and devices.

Comparison

User Mode	Kernel Mode
Limited privileges	Full privileges
No direct hardware access	Direct hardware access
Runs applications	Runs kernel
More secure	Highly privileged
Cannot execute privileged instructions	Can execute privileged instructions

Mode Switching

- 1 User program executes.
- 2 Program requests an OS service.
- 3 System call switches CPU to kernel mode.
- 4 Kernel performs the requested operation.
- 5 CPU returns to user mode.

Privileged Instructions

Examples include:

- Enable/Disable interrupts
- Configure memory management unit
- Access I/O ports
- Change processor control registers

These instructions execute only in kernel mode.

Airport Security

Passengers can access public areas only. Security staff can enter restricted areas.
Similarly:

- User Mode = Passenger
- Kernel Mode = Security Staff

Summary

- User mode protects the system from faulty applications.
- Kernel mode has complete hardware access.
- System calls switch execution from user mode to kernel mode.
- CPU returns to user mode after completing the service.

Quick Review

- ① Why are two CPU modes required?
- ② Differentiate user mode and kernel mode.
- ③ What is a privileged instruction?
- ④ Explain mode switching using a system call.